

LEDGER · 2026-08-27 · PLANT BIOTECHNOLOGY

High-Purity Shikonin via Aeroponic Hairy Root Bioreactors

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	98% Shikonin API derived from traditional field-grown Gromwell root (CAS 517-89-5)
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Entry application	nighttime wound-healing creams and anti-inflammatory serums
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Measured advantage	Product purity is identical to the incumbent at 98% API, and extraction yield improvements are only +20% (1.2x), failing the 2x performance superiority bar.
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Time to first revenue	18 months
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Gross margin at parity	74.2%
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Startup CapEx	\$325,000
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Payback from first sale	415.8 months
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Capital productivity

0.06x

Regulatory risk

Moderate — qualification costs reported (\$100,000 estimated)

Why it failed

- Voided by: Capital and time to revenue, Economics undisclosed, Negative control / sub-2x
- Margin fails the operating-cost check

Assessment

The venture fails commercial feasibility due to high startup CapEx (\$325,000), low gross margins (~41%), and regulatory delays pushing time to market past 18 months.

Volumetric biological limits constrain monthly production to ~3.42 kg, generating insufficient top-line revenue at \$450/kg market parity.

Also flagged

- Evidence rests on non-peer-reviewed sources
- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-27 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

biological or regulated processes that require appropriate expertise, facilities, permits and safety controls. Independently verify every figure, citation and regulatory requirement before committing capital.